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Master's Thesis in Informatics

Vision-Language Foundation Models for Multi-Modal Medical Data

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Vision-Language-Foundation-Modelle für Multimodale Medizinische Daten

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I confirm that this master's thesis is my own work and I have documented all sources and material used.

Munich, 16.03.2026 Muhammed Göktepe

Acknowledgments

Abstract

This thesis investigates organ-aware radiology report generation from 3D chest CT using vision-language foundation models. The main challenge is to preserve clinically relevant anatomical information while keeping the visual token interface compact and stable for language decoding.

We propose an end-to-end pipeline that combines: (i) volumetric CT preprocessing and organ-mask supervision, (ii) LLM-based decomposition of reports into organ-level targets, (iii) organ-conditioned visual token extraction from a 3D vision encoder, and (iv) decoder adaptation with parameter-efficient fine-tuning. The framework supports a unified comparison across MedGemma, GPT-2, and BioBART, with MedGemma-centered ablations completed in this thesis version.

Experiments on CT-RATE show that the strongest gains come from interface design rather than architectural complexity. The best MedGemma variant (`medgemma_lora_vis_token_pos_embed`) achieves the top global scores in key metrics, including GREEN (0.348), BLEU-4 (0.105), ROUGE-L (0.379), BERTScore (0.596), RadGraph (0.268), and RadCliQ (1.104), improving over the baseline in most clinically relevant dimensions. Perceiver and multiscale variants did not consistently outperform the simpler baseline under the same training budget, indicating that stable vision-language bridging is a higher-leverage design factor in this setting.

Overall, the thesis provides a reproducible organ-aware 3D report generation framework, code-grounded ablation evidence, and a practical direction for completing cross-decoder comparison with GPT-2 and BioBART under a shared evaluation protocol.

Contents

Acknowledgments **iv**

Abstract **v**

1 Introduction **1**
1.1 Problem Statement 1
1.2 Thesis Formulation 2
1.3 Contributions 2

2 Related Work **3**
2.1 Vision Transformers in Medical Imaging 3
2.1.1 Architectural Evolutions: Hierarchical and Self-Supervised ViTs 3
2.1.2 Adaptation to 3D Medical Volumes 4
2.2 Vision–Language Models 4
2.2.1 Contrastive Alignment and the CLIP Paradigm 4
2.2.2 Generative VLMs and Open-Weight Ecosystems 5
2.2.3 Medical Vision-Language Models 6
2.3 Decoders and Parameter-Efficient Fine-Tuning 7
2.3.1 Encoder-Decoder Architectures vs. Causal Modeling 7
2.3.2 Parameter-Efficient Fine-Tuning (PEFT) 8
2.4 Radiology Report Generation 9
2.4.1 Traditional and Early Transformer Approaches 9
2.4.2 The Evaluation Paradigm: From N-Grams to Clinical Efficacy . . 10
2.4.3 2D Radiology Foundation Models 10
2.4.4 3D Volumetric Report Generation 11
2.5 3D Medical Image Segmentation and Organ-Guided Attention 13
2.5.1 Volumetric Segmentation Frameworks 13
2.5.2 Organ-Guided Attention and Clinical Grounding 13

3 Methodology **15**
3.1 Problem Formulation and Pipeline Overview 15
3.2 Data Interfaces and Preprocessing 15
3.2.1 Volume and Mask Preprocessing 15

3.2.2 Target Organs and Segmentation-ID Mapping 16
3.2.3 Chest-CT Visibility Caveat 17
3.3 Report Decomposition for Organ-Level Supervision 17
3.3.1 LLM-Based Structuring 17
3.3.2 Prompted Supervision Format 17
3.4 Organ-Aware Visual Encoding 17
3.4.1 Backbone 17
3.4.2 Masked Cross-Attention with Organ Queries 18
3.4.3 Token Granularity: Multi-Token vs Compressed Tokens 18
3.4.4 Mask Safety and Empty-Organ Handling 18
3.5 Architecture Variants Used in Ablations 18
3.5.1 Baseline: ROI Cross-Attention + Projector 18
3.5.2 Positional Visual Prefix Variant 19
3.5.3 Perceiver Resampler Variant 19
3.5.4 Multi-Scale ViT-FPN Variant 19
3.5.5 MedGemma Architecture v3 Variant 19
3.6 Vision-Language Bridging 19
3.6.1 Projection and Stabilization 19
3.6.2 Sentinel Wrapping and Token Assembly 20
3.7 Decoder-Specific Adaptation 20
3.7.1 MedGemma Branch 20
3.7.2 GPT-2 and BioBART Branches 20
3.8 Training Objectives 21
3.8.1 Weighted Language Modeling Loss 21
3.8.2 Optional Alignment Loss 21
3.9 Optimization and Training Protocol 21
3.9.1 Shared Optimization Settings 21
3.9.2 Training-Strategy Ablations 22
3.10 Inference and Report Assembly 22
3.10.1 Organ-wise Generation 22
3.10.2 Structured Report Reconstruction 23
3.11 Model Comparison Matrix 23
3.12 Reproducibility and Implementation Notes 23

4 Experiments and Results **25**
4.1 Experimental Objectives and Research Questions 25
4.2 Dataset and Data Preparation 25
4.2.1 CT-RATE Overview 25
4.2.2 Preprocessing Protocol 25

4.2.3 Effective Organ Coverage in Evaluation 26
4.3 Experimental Setup 26
4.3.1 Implementation Stack 26
4.3.2 Common Optimization Settings 27
4.3.3 Model Families and Ablation Axes 27
4.4 Evaluation Metrics 27
4.4.1 Surface-level (Natural Language Generation) Metrics 29
4.4.2 Semantic Metric 30
4.4.3 Clinical and Structured Metrics 30
4.4.4 Metric Interpretation Policy 30
4.5 Automatic Evaluation Pipeline and Artifact Standardization 31
4.5.1 Pipeline 31
4.5.2 Metric Key Mapping 31
4.5.3 Source-of-Truth Policy for This Chapter 31
4.6 Results: Internal MedGemma Ablations 31
4.6.1 Primary Ablation Table 31
4.6.2 Full Variant Matrix 32
4.6.3 Relative Gains vs Baseline 32
4.6.4 Interpretation by Ablation Axis 32
4.7 Organ-Level Analysis 33
4.8 Cross-Attention Activation Case Studies 33
4.8.1 Case Study A: Typical Thoracic Pattern 34
4.8.2 Case Study B: Multi-Region Findings with Upper-Abdominal Context 34
4.8.3 Case Study C: Architectural Comparison (Baseline vs Perceiver/-Multiscale) 35
4.8.4 How to Read the Attention Maps 35
4.9 Decoder Comparison (Core Axis) 36
4.9.1 Main Comparison Table Structure 36
4.9.2 Current Run-Status Diagnostics (Before Final Metric Pass) 36
4.10 External Baseline: fVLM (Contextual) 37
4.11 Extended Ablation Protocols for Final Chapter Version 37
4.12 Threats to Validity and Interpretation Limits 38
4.13 Chapter Summary 38

5 Discussion **39**
5.1 Discussion Scope 39
5.2 Why Interface Design Was the Dominant Lever 39
5.2.1 Empirical Signal 39

5.2.2 Mechanistic Interpretation 39
5.2.3 Implication for 3D Medical VLM Design 40
5.3 Interpreting Negative or Mixed Ablations 40
5.3.1 Perceiver and Multiscale Variants 40
5.3.2 Curriculum and Sampling Strategies 40
5.4 Metric Disagreement and What to Trust 41
5.4.1 Why Disagreement Appears 41
5.4.2 CheXbert in Organ-Decomposed CT Text 41
5.4.3 Evaluation Policy Going Forward 41
5.5 Organ Coverage, Long-Tail Effects, and Generalization 41
5.6 Value of the Cross-Attention Case Studies 42
5.7 Decoder Axis: What Is Resolved and What Is Pending 42
5.8 Relation to Prior Paradigms in Medical Report Generation 42
5.9 Validity and Reporting Discipline 43
5.9.1 Internal Validity 43
5.9.2 External Validity 43
5.9.3 Construct Validity 43
5.10 Practical Implications 43
5.11 Chapter Synthesis 44

6 Conclusion and Future Work **45**
6.1 Conclusion 45
6.2 Take-Home Messages 45
6.3 Future Work 46
6.3.1 Track 1: Complete the Current Thesis Axis 46
6.3.2 Track 2: Improve Model Robustness and Efficiency 46
6.3.3 Track 3: Strengthen Clinical Validity 47
6.4 Final Remark 47

Abbreviations **48**

List of Figures **49**

List of Tables **50**

Bibliography **51**

1 Introduction

As deep learning has continued to advance over recent years, it has been applied to various fields of science and engineering. One of the most prominent applications of deep learning is in the field of computer vision. Deep learning models have achieved state-of-the-art performance on various computer vision tasks, such as image classification, object detection, and image segmentation. However, deep learning models are data-hungry and require large amounts of labeled data to achieve high performance. This is especially true for medical imaging, where obtaining large-scale labeled datasets is challenging due to privacy concerns and the high cost of expert annotation.

1.1 Problem Statement

Traditional NLP-based frameworks for radiology report generation often suffered from a lack of localized visual reasoning and a weak integration of clinical knowledge. While early 2D Vision-Language Models (VLMs) leveraging attention mechanisms improved the alignment between image regions and text tokens, they fundamentally failed to capture true three-dimensional anatomical context. In clinical practice, critical findings—such as lesion shape, spatial extent, and inter-slice continuity—depend entirely on volumetric data.

Applying modern VLM architectures directly to 3D medical volumes (like CT scans) introduces substantial computational bottlenecks. The cubic growth of 3D attention mechanisms requires bridging architectures, such as perceptual resamplers or Q-Formers, to compress thousands of spatial tokens into a dense semantic representation without overwhelming the language decoder. Furthermore, while the field of generative AI has seen an expansion of capable multi-billion parameter foundation models, it remains an open question whether accurate medical report generation depends primarily on the 3D vision encoder or on the semantic reasoning capacity of the Large Language Model decoder.

1.2 Thesis Formulation

This thesis addresses these gaps by evaluating state-of-the-art 3D Vision-Language Modeling techniques for radiology report generation. Specifically, we investigate whether extracting single, highly semantic visual tokens per anatomical organ provides a cleaner signal to the LLM decoder than complex, multi-token representations.

To achieve this, we leverage an organ-guided decomposition methodology where reports are broken down anatomical region by region, enabling the VLM to perform targeted, localized reasoning. Our experiments are anchored on the **CT-RATE** dataset [1], a large-scale public dataset pairing 3D chest CT volumes with free-text radiology reports.

Instead of relying solely on traditional natural language metrics (BLEU, ROUGE) which often fail to capture clinical nuances—such as severely penalizing negative findings—this work assesses model performance using advanced clinical efficacy metrics like **RadGraph** [2] and **GREEN** (Generative Radiology Report Evaluation and Error Notation) [3], ensuring generated reports are clinically actionable rather than linguistically mimetic.

By running extensive ablations across decoder architectures—comparing medically aligned causal language models (e.g., **MedGemma** [4]) against generic causal models (**GPT-2**) and encoder-decoder architectures (**BioBART**)—this research analyzes how decoder choice affects mapping sparse 3D visual tokens into structured clinical text.

1.3 Contributions

The main contributions of this thesis are as follows:

1. **Comprehensive 3D VLM Benchmarking**: We conduct one of the first extensive comparative analysis of decoder architectures (causal vs. encoder-decoder) coupled with 3D vision encoders for volumetric report generation.

2. **Organ-Guided Tokenization Strategy**: We show that compressing high-density 3D spatial features into a single, organ-specific visual token (via cross-attention) outperforms more complex multi-token bridging architectures in our current ablations.

3. **Clinical Efficacy Prioritization**: We validate generated results strictly against clinical-factuality metrics (GREEN, RadGraph) to isolate true clinical reasoning from rote n-gram memorization.

2 Related Work

This chapter reviews the key areas of prior work that form the foundation for this thesis. Section 2.1 discusses the Vision Transformer architecture and its adaptation to medical imaging. Section 2.2 covers vision–language models, including both general-purpose and medical variants. Section 2.3 introduces large language models and parameter-efficient fine-tuning techniques. Section 2.4 surveys radiology report generation methods. Finally, Section 2.5 covers 3D medical image segmentation and organ-guided attention mechanisms.

2.1 Vision Transformers in Medical Imaging

The Vision Transformer (ViT) [5] fundamentally shifted the paradigm of computer vision by applying the transformer architecture, originally designed for natural language processing, to image recognition. ViT divides an input image into fixed-size non-overlapping patches, projects each patch into an embedding, and processes the resulting sequence through a standard transformer encoder. By treating image patches as tokens and relying entirely on multi-head self-attention, ViT eliminates the need for convolutional inductive biases (such as translation invariance and local receptive fields) and achieves competitive or superior performance compared to Convolutional Neural Networks (CNNs) when trained on massive datasets like ImageNet.

2.1.1 Architectural Evolutions: Hierarchical and Self-Supervised ViTs

While the original ViT required large-scale supervised pretraining, subsequent architectures addressed its limitations. The Swin Transformer [6] introduced a hierarchical structure with shifted windows, computing self-attention locally within non-overlapping windows that shift across layers. This hierarchical design reintroduces some useful inductive biases, making it highly efficient for dense prediction tasks like segmentation and object detection, which are critical in medical imaging contexts where precise anatomical boundaries matter.

Simultaneously, advancements in self-supervised learning alleviated the reliance on massive labeled datasets—a severe bottleneck in the medical domain. Masked Autoencoders (MAE) [7] proposed an asymmetric encoder-decoder architecture where the

encoder operates only on an unmasked subset of patches (e.g., 25%), and a lightweight decoder reconstructs the heavily masked (75%) original image. This approach enables the learning of robust, high-level visual representations directly from unlabeled data. Building upon self-supervised paradigms, DINOv2 [8] demonstrated that models trained via self-distillation with no labels can learn universal visual features that are highly transferable across diverse domains, capturing part-level semantic correspondences without explicit supervision.

2.1.2 Adaptation to 3D Medical Volumes

In the medical imaging domain, transformers have been widely adopted to capture the long-range dependencies crucial for understanding complex anatomical structures. Early adaptations like TransUNet [9] combined a CNN encoder for local feature extraction with a transformer bottleneck for global context in 2D medical image segmentation.

However, clinical data such as Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) are inherently 3D volumetric scans. Processing these volume slice-by-slice discards critical through-plane spatial context. UNETR [10] addressed this by using a ViT encoder to process 3D volumetric data directly, extracting 3D patches from the input volume.

The transition from 2D image patches to 3D volumetric patches introduces substantial computational cost. In a standard self-attention mechanism, memory and computational complexity grow quadratically with sequence length. For a 3D medical image, sequence length expands cubically with respect to spatial dimensions. If a high-resolution CT scan is embedded into thousands of 3D tokens, global self-attention can exceed the memory limits of modern GPUs. To mitigate this, modern 3D Vision Transformers commonly use sparse attention, shifted-window variants, or early spatial downsampling.

2.2 Vision–Language Models

Vision–language models (VLMs) aim to bridge the semantic gap between visual perception and natural language understanding. These models learn joint representations of images and text, enabling a wide array of cross-modal tasks such as image captioning, visual question answering (VQA), and complex multi-modal reasoning.

2.2.1 Contrastive Alignment and the CLIP Paradigm

CLIP (Contrastive Language–Image Pre-training) [11] introduced a highly scalable approach to learning visual representations from natural language supervision. By training independent vision and text encoders to maximize the cosine similarity of

matching image–text pairs in a shared embedding space while minimizing the similarity of millions of non-matching pairs, CLIP learns highly transferable visual concepts. Trained on 400 million image–text pairs curated from the internet, CLIP demonstrates strong zero-shot transfer capabilities across a wide range of downstream visual tasks.

Recent advances have refined the CLIP objective. SigLIP (Sigmoid Loss for Language Image Pre-Training) [12] replaces the standard softmax-based contrastive loss with a pairwise sigmoid loss. This change removes the need for global views of all negative pairs across devices, improving training efficiency and scalability and often yielding stronger representations. The contrastive learning paradigm established by CLIP and extended by SigLIP remains a common foundation for aligning vision and language modalities in modern VLMs.

2.2.2 Generative VLMs and Open-Weight Ecosystems

While contrastive models focus purely on alignment and retrieval, generative VLMs extend the paradigm to produce natural language outputs explicitly conditioned on visual inputs. BLIP (Bootstrapping Language-Image Pre-training) [13] proposed a unified framework that jointly trains a contrastive alignment objective, an image–text matching classification objective, and a generative language modeling objective. This multi-task design enables BLIP to be highly effective for both comprehensive image understanding and robust text generation tasks.

Building upon this, BLIP-2 [14] introduced a compute-efficient strategy for bridging the modality gap: the Q-Former (Query Transformer). The Q-Former acts as an informational bottleneck between a frozen image encoder and a frozen large language model. It utilizes a set of learnable query vectors that cross-attend to the dense visual features extracted by the image encoder, forcing the queries to extract only the most text-relevant visual information. By doing so, the Q-Former compresses thousands of spatial visual tokens into a fixed, small number of semantic latent tokens (e.g., 32 tokens). These compressed tokens are then linearly projected directly into the text embedding space of the frozen LLM. This architecture demonstrated that one can achieve strong multi-modal capabilities without unfreezing the massive weights of the underlying language or vision foundation models, reducing computational cost.

A similar design philosophy for handling high-density visual signals was adopted by DeepMind's Flamingo [15]. Rather than a Q-Former, Flamingo utilized a Perceiver Resampler mechanism, which also cross-attends to a large input array (such as a high-resolution image or a varying-length video) using a small set of latent, learnable query vectors. Flamingo then interleaved these compressed visual tokens with standard text tokens inside a frozen LLM through gated cross-attention layers, demonstrating strong few-shot learning and in-context reasoning capabilities.

LLaVA (Large Language-and-Vision Assistant) [16] vastly simplified these complex bridging architectures by proving that a straightforward Multi-Layer Perceptron (MLP) projection layer could effectively map visual features directly from a pretrained CLIP encoder into the native vocabulary embedding space of an LLM, entirely discarding the need for Q-Formers or Resamplers. Through visual instruction tuning—training the model on instruction-following datasets combining images with conversational text—LLaVA achieved remarkable multi-modal dialogue capabilities. Recently, open-source initiatives have rapidly accelerated VLM research. Models like InternVL [17] focus on massive scaling of the vision foundation model to better align with generic linguistic tasks, while the Molmo family [18] emphasizes the crucial role of highly curated, completely open-source point-based multimodal data to achieve state-of-the-art results with highly efficient projector architectures.

2.2.3 Medical Vision-Language Models

Adapting VLMs to the medical domain presents significant unique challenges. Medical images (such as high-resolution multi-parametric MRIs or chest X-rays) have fundamentally different characteristics from natural images (like those in COCO/ImageNet). Furthermore, clinical texts utilize a dense, highly specialized vocabulary, and tolerance for error in medical AI is low.

Early attempts to bridge this gap focused on domain-specific contrastive pretraining. BiomedCLIP [19] addressed this by pretraining a CLIP-style architecture from scratch on 15 million biomedical image–text pairs sourced from PubMed Central articles, demonstrating that in-domain pretraining improves performance on downstream biomedical classification and retrieval tasks compared to generic CLIP models.

For 2D clinical radiology, development has accelerated in recent years [20]. CheXagent [21] was introduced as a clinical foundation model for chest X-ray interpretation, fine-tuned on large corpora of radiology reports to interpret images and answer clinical questions. MAIRA-2 [22] advanced grounded report generation by linking generated assertions to localized bounding boxes in the original X-ray, reducing hallucinations and improving trustworthiness.

For 3D medical imaging, the challenge is compounded by the sheer size of the volumetric data and the scarcity of paired text data. Hamamci et al. [1] introduced CT-CLIP, a contrastive learning framework specifically engineered for 3D computed tomography. Trained exclusively on the massive CT-RATE dataset—which pairs unannotated 3D volumetric CT scans directly with their corresponding free-text radiology reports—CT-CLIP successfully learns 3D structural visual representations that map directly to the clinical text manifold. This foundational work represents one of the first successful large-scale efforts to apply multi-modal vision–language pretraining directly

to unstructured volumetric medical data.

MedGemma [4] is a family of medical foundation models built on top of Google's rigorous Gemma architecture [23]. MedGemma structurally inherits the robust capabilities of the general-purpose Gemma model but undergoes rigorous continued pretraining and specialized instruction-tuning on vast troves of verified biomedical literature, medical board exams, and de-identified clinical notes. The resulting instruction-tuned variant (MedGemma-4B-IT) demonstrates exceptional zero-shot and few-shot capability on complex medical reasoning tasks, making it a highly compelling, medically-aligned language decoder for complex medical VLM pipelines, and ultimately serving as one of the core generative components evaluated in this thesis.

2.3 Decoders and Parameter-Efficient Fine-Tuning

The generation of complex radiology reports from visual features requires a demonstrably powerful language decoder. The development of natural language generation models has largely bifurcated into two fundamentally distinct architectural paradigms: Encoder-Decoder models and Causal Language Models. Understanding the mechanistic differences between these paradigms is crucial when selecting a backbone for medical vision-language reasoning.

2.3.1 Encoder-Decoder Architectures vs. Causal Modeling

Encoder-Decoder models, such as BART (Bidirectional and Auto-Regressive Transformers) [24] and T5, utilize a bidirectional encoder to process the input context comprehensively and an autoregressive decoder to generate the output sequence token by token. Because the encoder can attend to both left and right context simultaneously during the prompt processing phase, these models excel at tasks requiring deep semantic understanding of the entire input condition before generating text (e.g., summarization or machine translation). BART is specifically pretrained by corrupting documents (via token masking, deletion, and sentence permutation) and optimizing a reconstruction loss.

In the biomedical domain, BioBART [25] adapts this exact architecture by continually pretraining the baseline BART model on a massive corpus of PubMed abstracts. This domain-specific pretraining allows BioBART to inherit the robust grammatical generation capabilities of BART while rigidly aligning its vocabulary and internal latent representations with complex, highly specialized medical terminology. In standard vision-language pipelines utilizing Encoder-Decoder backbones, visual embeddings are typically fed directly into the bidirectional encoder, allowing the system to build a rich,

globally aware contextual representation before the autoregressive generation phase even begins.

Conversely, Causal Language Models (CLMs) utilize a decoder-only architecture that predicts the next token in a sequence strictly based on preceding leftward tokens. GPT-2 (Generative Pre-trained Transformer 2) [26] popularized this highly scalable paradigm, demonstrating that simple unsupervised next-token prediction on diverse web corpora could intuitively yield models capable of coherent, zero-shot, open-ended text generation. Due to its relatively modest scale (e.g., 124M parameters for the base model) and exceptionally clean architectural foundation, GPT-2 remains a popular, tractable baseline for experimental diagnostic text generation tasks where compute is constrained.

Unlike bidirectional encoders, causal models process the conditioning prompt (including visual tokens prepended as a "soft prompt") autoregressively. While theoretically limiting the deep bidirectional mingling of context tokens, the massive parameter count of modern causal models overrides this limitation through sheer representational capacity.

Recent foundation models have scaled the decoder-only paradigm to tens of billions of parameters. The LLaMA family [27] and the Gemma family [23] have demonstrated strong capabilities in generalized natural language understanding, logical reasoning, and instruction-following. MedGemma [4], a medical variant (e.g., MedGemma-4B-IT), extends these capabilities with continued medical pretraining and instruction tuning. In causal VLM architectures, preserving the spatial and semantic integrity of the visual token prefix is important because downstream generation depends directly on this initial visual context.

2.3.2 Parameter-Efficient Fine-Tuning (PEFT)

Adapting massive decoders (especially multi-billion parameter foundation models like MedGemma) to specialized downstream tasks such as highly localized, organ-specific radiology report generation via full fine-tuning is computationally prohibitive and can degrade the base model's general linguistic capabilities. Parameter-efficient fine-tuning (PEFT) methods address this by updating only a small fraction of the model's parameters while keeping most pretrained weights frozen.

Early PEFT strategies included Adapter modules [28], which introduced small, trainable feed-forward bottleneck networks sandwiched between the frozen multi-head attention and feed-forward transformer sub-layers. Another approach, Prefix-Tuning [29], mathematically optimizes a sequence of continuous task-specific prompt vectors prepended to the transformer layers. While effective in reducing forgetting, these methods alter the network topology, often introducing additional inference latency

or reducing the available context window depth.

LoRA (Low-Rank Adaptation) [30] addresses these issues and is now widely adopted in foundation-model fine-tuning. LoRA assumes that required updates to dense weight matrices have low intrinsic rank. It introduces trainable low-rank matrices in parallel with frozen linear layers (commonly query and value projections in self-attention). Instead of updating the full matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA learns smaller matrices $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times k}$ where $r \ll \min(d, k)$, yielding $W_0 + \Delta W = W_0 + BA$. Because BA can be merged into W_0 for deployment, LoRA does not add extra inference latency after merging.

To further reduce training cost, QLoRA [31] combines LoRA with 4-bit Normal Float (NF4) double quantization. In QLoRA, the frozen pretrained model weights reside in 4-bit precision, while only the lightweight, trainable LoRA adapter matrices are cast to higher precision (e.g., BFloat16) during backpropagation. This combination substantially reduces VRAM requirements for fine-tuning multi-billion parameter foundation models, allowing faster iteration on standard hardware.

2.4 Radiology Report Generation

Automatic radiology report generation is the task of producing clinically accurate free-text reports from medical images. This task is fundamentally more challenging than generic image captioning because generated reports must be factually accurate, cover all relevant pathological findings (and explicitly note the absence of others), and strictly adhere to dense clinical conventions.

2.4.1 Traditional and Early Transformer Approaches

Early approaches to radiology report generation adapted encoder–decoder architectures directly from the natural image captioning literature. Jing et al. [32] proposed a co-attention mechanism combined with a hierarchical LSTM decoder to generate structured medical reports from chest X-rays. However, these CNN–RNN frameworks typically produced generic, repetitive descriptions and struggled significantly with capturing rare or subtle findings, often suffering from severe data imbalance in clinical datasets.

Transformer-based methods subsequently improved report generation quality by better tracking long-term context. Chen et al. [33] introduced a memory-driven transformer that used a relational memory module to record key clinical information during autoregressive generation, markedly improving the coherence of long reports. In a follow-up, Chen et al. [34] proposed cross-modal memory networks that maintained separate but interacting memory banks for visual and textual features, enabling more robust cross-modal reasoning. Demonstrating the value of initialization, Nicolson et

al. [35] showed that warm-starting both the vision encoder (e.g., from ImageNet or CheXpert) and the language decoder (e.g., from clinical BERT) substantially improved chest X-ray report generation over training from scratch.

2.4.2 The Evaluation Paradigm: From N-Grams to Clinical Efficacy

A foundational obstacle in early radiology report generation research was the over-reliance on traditional Natural Language Generation (NLG) metrics originally designed for machine translation—such as BLEU, ROUGE, and METEOR. These metrics inherently quantify lexical overlap; they measure how many identical n-grams the generated text shares with the ground truth. While effective for general captioning, they are critically flawed in the medical domain. A single word substitution (e.g., swapping "no" for "a") has a minimal penalty on a BLEU score but completely reverses the clinical diagnosis of the report, potentially causing severe harm in a deployed setting.

To address this gap, the research community shifted toward metrics that explicitly measure clinical efficacy and semantic correctness. Early clinical metrics like CheXbert [36] leveraged an NLP labeler to extract predefined clinical observations (e.g., Pneumonia, Cardiomegaly) from both the generated and target reports, scoring the model based on diagnostic accuracy rather than exact wording.

Subsequent innovations focused on capturing the relational structure of medical language. The RadGraph metric [2] evaluates reports by parsing them into knowledge graphs (nodes representing entities like anatomy or observation, and edges representing relations like "located at" or "modify"), computing overlap between structured clinical graphs. More recently, the GREEN (Generative RadiOlogy Report Evaluation and Error aNnotation) metric [3] provides both a quantitative clinical score and qualitative feedback that correlates well with expert judgment. This shift in evaluation metrics moved model design away from pure lexical matching toward factual and clinically grounded generation.

2.4.3 2D Radiology Foundation Models

As the field moved toward the foundation model paradigm, researchers began adapting large, frozen language models as report decoders. R2GenGPT [37] demonstrated that a frozen LLM combined with a trainable visual encoder and a lightweight projection layer could generate high-quality radiology reports with competitive performance on both NLG and clinical efficacy metrics. This paradigm benefits directly from the language understanding capabilities and medical knowledge embedded in pretrained LLMs while keeping multi-modal training costs manageable.

Recently, the field has seen the emergence of true foundation models specifically

tailored for 2D X-ray interpretation. CheXagent [21] stands out as a pioneering clinical foundation model designed to interpret chest X-rays. By fine-tuning a massive VLM on diverse corpora of radiology reports and visual question-answering pairs, CheXagent achieved state-of-the-art performance in both free-text report generation and interactive clinical dialogue.

Simultaneously, the architectural focus shifted from mere linguistic fluency to rigorous clinical grounding. MAIRA-2 [22] introduced the concept of grounded radiology report generation. Instead of hallucinating plausible text based on global image context, MAIRA-2 explicitly maps generated clinical assertions in the text back to localized bounding boxes in the original X-ray. This spatial grounding dramatically reduces clinical hallucinations—a critical safety requirement for medical deployment.

2.4.4 3D Volumetric Report Generation

While 2D X-ray interpretation has matured rapidly, 3D volumetric report generation (e.g., from CT or MRI scans) remains an open and highly challenging frontier. The jump from 2D to 3D exponentially increases the computational complexity of the visual token sequence. Furthermore, until the recent release of the CT-RATE dataset [1], there was a severe scarcity of paired 3D volumes and free-text reports available for training.

The release of CT-RATE spurred the rapid development of several pioneering 3D multimodal foundation models:

- **CT2Rep** [38]: One of the earliest works to leverage the CT-RATE dataset, CT2Rep demonstrated that automated radiology report generation for 3D medical imaging is feasible by properly projecting 3D visual tokens into a robust language decoder.
- **M3D** [39]: Advancing 3D medical image analysis with Multi-Modal Large Language Models, M3D proposed strategies to efficiently compress volumetric data into spatial tokens, enabling the LLM to process entire CT scans without exceeding context windows.
- **Argus** [40]: Argus introduced comprehensive benchmarking and enhancement strategies for VLMs in 3D radiology, testing models ranging from 3B configurations to massive 70B parameter deployments. Argus highlighted that scaling the LLM capacity yields direct improvements in clinical efficacy metrics like GREEN.
- **Med3DVLM** [41] and **RadFM** [42]: Both models established unified 3D vision-language frameworks, focusing on cross-modal alignment between highly unstructured 3D voxel spaces and precise clinical text.

Recent work has also emphasized fine-grained anatomical supervision for CT understanding. fVLM [43] introduces large-scale and fine-grained vision-language pretraining for CT and reports strong gains on report generation benchmarks. Its organ-aware supervision and decomposition perspective directly motivate the organ-guided decomposition strategy used in this thesis.

A key challenge unifying these works is token compression. Because a raw 3D CT volume contains vastly more information than a 2D image, naive patch extraction results in tens of thousands of tokens that can overwhelm the LLM decoder. Recent studies [44] emphasize that tokenization strategies (such as Perceiver-based latent resampling, spatial 3D pooling, or learned region-of-interest selection) are an important factor for effective 3D VLM performance.

Table 2.1: Performance benchmarking of recent 3D Volumetric Foundation Models for Radiology Report Generation evaluated on the CT-RATE dataset. Metrics highlight the paradigm shift toward clinical efficacy (GREEN) over traditional NLP metrics (BLEU-4). Data adapted from recent multi-modal surveys.

Model Name	**Parameters**	**Input Resolution**	**BLEU-4**	**ROUGE-L**	**METEOR**	**GREEN**
Argus-70B	70B	$512 \times 512 \times 256$	47.98	-	-	**60.22**
Argus-8B	8B	$512 \times 512 \times 256$	45.81	-	-	58.42
Argus-3B	3B	$512 \times 512 \times 256$	43.03	-	-	55.56
CT2Rep-3B	3B	$256 \times 256 \times 64$	38.77	48.96	-	51.72
M3D-3B	3B	$256 \times 256 \times 64$	40.23	50.31	-	50.31
RadFM-3B	3B	$256 \times 256 \times 64$	39.95	49.46	-	49.46
COLIPRI-CRM	1B	$160 \times 160 \times 160$	45.20	55.20	57.80	-

As shown in Table 2.1, under the reported experimental settings, higher resolution inputs and larger language decoders (e.g., Argus-70B) are associated with higher GREEN scores.

The approach adopted in this thesis builds directly upon this frontier: a trainable 3D vision encoder (with variants exploring multi-scale and organ-guided context) produces visual embeddings that are projected specifically into the embedding space of a medically aligned MedGemma decoder, targeting the complex transition from global captioning to precise, localized, organ-specific 3D reporting.

2.5 3D Medical Image Segmentation and Organ-Guided Attention

Accurate anatomical localization is an essential prerequisite for generating highly specific, clinically actionable radiology reports. The availability of robust, automated whole-body 3D segmentation tools has enabled novel VLM architectures that explicitly leverage anatomical priors to ground their text generation.

2.5.1 Volumetric Segmentation Frameworks

While early segmentation models relied on 3D CNNs (e.g., 3D U-Net), transformer-based architectures have increasingly dominated this space by effectively capturing long-range contextual dependencies. TransUNet [9] and Swin-UNet [45] demonstrated the efficacy of transformers for 2D medical segmentation, but volumetric data requires dedicated 3D processing.

nnFormer (Interleaved Transformer for Volumetric Medical Image Segmentation) [46] extends transformer capabilities explicitly to 3D medical images. By interleaving local 3D convolution operations with global multi-head self-attention mechanisms, nnFormer captures both the intricate local textures of tumors and the global geometrical relationships of large organs, setting strong benchmarks across multiple 3D medical datasets.

In practical clinical pipelines, TotalSegmentator [47] has emerged as an indispensable, widely used utility for automatic segmentation. Built upon the robust nnU-Net framework, TotalSegmentator provides highly reliable, out-of-the-box segmentation of 104 distinct anatomical structures—including major organs, bones, vessels, and muscles—from standard CT images. The discrete 3D masks produced by such algorithms can serve directly as precise anatomical priors for downstream generative tasks.

2.5.2 Organ-Guided Attention and Clinical Grounding

Incorporating these anatomical priors into VLM architectures enables a shift from global image captioning to localized, organ-specific reasoning. Rather than processing an entire CT volume as a uniform sequence of visual tokens, organ-guided mechanisms restrict or bias the model's cross-attention to regions corresponding to specific query structures (e.g., the lungs or the liver).

This approach draws heavily upon learnable query architectures such as the Perceiver [48], where a fixed set of latent vectors attends to a massive sensory input. In the medical domain, these latent vectors can represent specific organs. By masking the visual features using 3D segmentation maps prior to cross-attention, the language decoder

is forced to focus only on the relevant anatomy when generating the corresponding sentence in the radiology report.

Furthermore, in the context of multi-task learning [49], organ-specific feature extraction can be significantly enhanced via auxiliary classification objectives. By jointly training the visual encoder to perform specific disease classification tasks alongside open-ended report generation, the model learns a more disentangled and discriminative organ-level representation. This auxiliary grounding acts as a powerful inductive bias, encouraging the visual encoder to capture the subtle physiological abnormalities that drive accurate clinical reporting.

3 Methodology

3.1 Problem Formulation and Pipeline Overview

This thesis studies organ-aware report generation from 3D chest CT. The model input is a CT volume $V \in \mathbb{R}^{D \times H \times W}$, together with organ masks $\{M_o\}_{o \in \mathcal{O}}$ for a fixed organ set $\mathcal{O}$. The output is a structured report where each organ receives a dedicated text segment, and all segments are assembled into a final report for evaluation.

Formally, for patient i, we model:

$$\hat{Y}_i = f_\theta(V_i, \{M_{i,o}\}_{o \in \mathcal{O}}), \tag{3.1}$$

where $\hat{Y}_i = \{\hat{y}_{i,o}\}_{o \in \mathcal{O}}$ contains organ-level generated text. We then build a report-level output by concatenating organ sections in a fixed anatomical order.

The end-to-end pipeline contains six stages:

1. CT preprocessing and mask preparation.
2. LLM-based decomposition of reference reports into organ-level supervision.
3. Organ-aware visual token extraction from a 3D ViT backbone.
4. Vision-language bridging into decoder embedding space.
5. Organ-conditioned autoregressive generation.
6. Organ-section assembly and report-level evaluation.

The same overall interface is used for three decoder families: MedGemma [4], GPT-2 [26], and BioBART [25]. This keeps decoder comparison meaningful by controlling visual input structure and data protocol.

3.2 Data Interfaces and Preprocessing

3.2.1 Volume and Mask Preprocessing

The training and evaluation code uses a consistent MONAI-based transform stack across model families. Given raw NIfTI CT and segmentation volumes, preprocessing applies:

1. channel standardization and axis transposition,
2. HU intensity clipping ($[-1150, 350]$) and min-max scaling to $[0, 1]$,
3. spatial padding to $(112, 256, 352)$,
4. final resize to $(112, 256, 352)$ with trilinear interpolation for CT and nearest-neighbor for masks.

This choice intentionally replaces center-cropping with resizing to reduce accidental removal of peripheral anatomy (especially small or partially visible structures).

3.2.2 Target Organs and Segmentation-ID Mapping

The final training schema uses 12 target organs that are sufficiently represented in the decomposed reports and compatible with available segmentation labels:

$$\mathcal{O} = \{\text{lung, heart, esophagus, liver, gallbladder, stomach, pancreas, spleen, kidney, aorta, trachea, rib}\}. \tag{3.2}$$

Mask extraction in data loaders follows fixed integer-ID mappings. Table 3.1 summarizes the mapping used in training and evaluation scripts.

Table 3.1: Organ-to-segmentation-ID mapping used for binary mask construction.

Organ	**Segmentation IDs**
Lung	10, 11, 12, 13, 14
Heart	51, 61
Aorta	52
Esophagus	15
Trachea	16
Rib	92–115
Liver	5
Gallbladder	4
Stomach	6
Pancreas	7
Spleen	1
Kidney	2, 3

3.2.3 Chest-CT Visibility Caveat

Although the task is chest CT, several upper-abdominal organs are retained because they appear in a non-trivial fraction of scans. However, visibility is inconsistent across subjects and protocols. Therefore, organs with limited field-of-view support are treated as optional supervision targets rather than mandatory textual outputs.

3.3 Report Decomposition for Organ-Level Supervision

3.3.1 LLM-Based Structuring

Reference reports are decomposed into organ-keyed JSON records using an instruction-following LLM pipeline (Llama-3.3-70B-Instruct through vLLM [50], with AWQ quantization [51]). The pipeline enforces:

1. strict schema-constrained JSON output,
2. one textual value per organ key,
3. explicit handling of missing organs as empty/null values.

This decomposition converts free-text report supervision into organ-conditioned generation targets and enables direct organ-level error analysis.

3.3.2 Prompted Supervision Format

During training, each organ is paired with an organ-specific instruction prompt. In MedGemma branches, chat-format prompts are used:

```
<start_of_turn>user ...  Describe the findings for the [organ] ...
<start_of_turn>model
```

For GPT-2/BioBART branches, a shorter prompt style is used (e.g., `Describe [organ]:`).

The loss is masked to the response region only; prompt tokens and padding tokens are ignored.

3.4 Organ-Aware Visual Encoding

3.4.1 Backbone

All main variants use a 3D ViT encoder initialized from a pretrained checkpoint and fine-tuned end-to-end. The standard configuration operates on $(112, 256, 352)$ volumes with patch size $(16, 16, 32)$ and hidden size 768.

Some ablations modify this backbone (e.g., CNN stem before ViT, multiscale fusion), but the output interface to the language side remains organ-conditioned tokens.

3.4.2 Masked Cross-Attention with Organ Queries

Let $F(V) \in \mathbb{R}^{T \times d_v}$ be ViT features and $Q_o \in \mathbb{R}^{K \times d_v}$ be K learned queries for organ o. Binary organ masks are downsampled to the ViT token grid and converted to an additive attention bias matrix B_o, where background tokens receive $-\infty$.

Organ token extraction uses masked cross-attention:

$$Z_o = \mathrm{MHA}(Q_o, F(V), F(V); B_o), \tag{3.3}$$

where $Z_o \in \mathbb{R}^{K \times d_v}$ and K is the token count per organ.

In the implementation, the downsampled mask size is fixed to $(7, 16, 11)$ to match the ViT tokenization geometry used in training.

3.4.3 Token Granularity: Multi-Token vs Compressed Tokens

A major design axis is token granularity per organ:

- **Multi-token mode:** $K = 8$ visual tokens per organ (96 total for 12 organs).
- **Compressed mode:** one token per organ after aggregation/projection (effective $K = 1$ at decoder interface).

The compressed strategy is motivated by 3D token-budget constraints and empirical instability when long visual prefixes are combined with long text prompts.

3.4.4 Mask Safety and Empty-Organ Handling

A practical issue in organ-conditioned attention is empty masks. If all attention positions are blocked (all $-\infty$), softmax becomes undefined. The implementation detects this case and resets the bias to zeros for that organ-query path, allowing fallback to unconstrained attention rather than producing NaNs.

3.5 Architecture Variants Used in Ablations

3.5.1 Baseline: ROI Cross-Attention + Projector

The baseline branch uses one masked cross-attention stage over ViT features and a linear projector into decoder embedding space.

3.5.2 Positional Visual Prefix Variant

The best-performing MedGemma variant adds learned visual positional embeddings and explicit sentinel wrapping (`<vis>` and `<end_vis>`) around visual tokens before text tokens. This helps the decoder separate visual context from instruction text.

3.5.3 Perceiver Resampler Variant

A Perceiver-style resampler [48] replaces single-step cross-attention with multiple layers, each containing:

1. self-attention among organ queries,
2. cross-attention from queries to ViT features,
3. feed-forward refinement.

This increases modeling capacity but also compute and optimization complexity.

3.5.4 Multi-Scale ViT-FPN Variant

The multi-scale variant fuses intermediate ViT block features (early/mid/late/final) using learned scale weights and a fusion projection. The goal is to preserve both coarse and fine signals, especially for small or low-contrast structures.

3.5.5 MedGemma Architecture v3 Variant

The v3 branch introduces a CNN stem before ViT, changes the ViT patch geometry accordingly, and adds a text-visual alignment term during training. This variant explores whether stronger inductive bias in the encoder and auxiliary contrastive alignment improves organ grounding.

3.6 Vision-Language Bridging

3.6.1 Projection and Stabilization

Visual features are projected from ViT dimension $d_v = 768$ to decoder hidden size d_l:

$$\tilde{Z}_o = W_p Z_o + b_p, \quad W_p \in \mathbb{R}^{d_v \times d_l}. \tag{3.4}$$

Two stabilization operations are applied before decoding:

1. layer normalization on projected visual embeddings,

2. variance matching to decoder token embedding scale.

The variance matching step reduces "embedding shock" when concatenating projected visual vectors with pretrained language embeddings.

3.6.2 Sentinel Wrapping and Token Assembly

For MedGemma, the effective input sequence is:

$$[\text{BOS}],\ \texttt{<vis>},\ \tilde{Z}_o,\ \texttt{<end_vis>},\ \text{text prompt and target tokens.} \tag{3.5}$$

Loss is masked over the inserted visual block so that only clinically relevant text tokens contribute to language-modeling gradients.

3.7 Decoder-Specific Adaptation

3.7.1 MedGemma Branch

MedGemma is loaded as a frozen base decoder, then adapted with LoRA [30]. LoRA settings used in main runs are:

- rank $r = 16$,
- $\alpha = 32$,
- dropout 0.05,
- target modules: `q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj`.

This keeps training memory manageable while allowing meaningful decoder adaptation under organ-conditioned visual inputs.

3.7.2 GPT-2 and BioBART Branches

For GPT-2/BioBART, the model is wrapped in a `VisionEncoderDecoderModel` interface. The encoder output is organ-token embeddings; decoder-side adaptation unfreezes a restricted subset of parameters (cross-attention blocks, normalization layers, output head, and embeddings) while keeping most pretrained weights frozen.

For seq2seq decoders (BioBART), pretrained shared embeddings and output head weights are explicitly copied from the original seq2seq checkpoint to avoid random initialization artifacts in causal-decoding wrappers.

3.8 Training Objectives

3.8.1 Weighted Language Modeling Loss

Given batch index i, organ o, and token position t, the primary loss is weighted token-level cross-entropy:

$$\mathcal{L}_{\text{LM}} = \frac{1}{B|\mathcal{O}|} \sum_{i=1}^{B} \sum_{o \in \mathcal{O}} w_{i,o} \left(\frac{1}{T_{i,o}} \sum_{t=1}^{T_{i,o}} \text{CE}(y_{i,o,t}, \hat{y}_{i,o,t}) \right), \tag{3.6}$$

where $w_{i,o}$ is an organ-sample weight used for imbalance handling.

The weighting mechanism differs across ablations:

- probability-based undersampling of default "no finding" targets,
- explicit organ loss weights from precomputed statistics,
- curriculum or hard-example dynamic weights.

3.8.2 Optional Alignment Loss

Some variants (notably `medgemma_architecture_v3` and GPT-2/BioBART branches) add a contrastive alignment objective between organ visual and organ text representations. For patient-level organ sets, similarity logits are computed as:

$$S_{o,o'} = \frac{\langle v_o, t_{o'} \rangle}{\tau}, \tag{3.7}$$

followed by cross-entropy over matching organ indices.

The total objective becomes:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda \mathcal{L}_{\text{align}}, \tag{3.8}$$

with λ set by variant-specific training code.

3.9 Optimization and Training Protocol

3.9.1 Shared Optimization Settings

Across major runs, training arguments are largely standardized:

- batch size: 1 per device,
- gradient accumulation: 8,

- learning rate: 1×10^{-4},
- weight decay: 0.01,
- warmup ratio: 0.05,
- epochs: 2 (retraining phase),
- bf16 enabled, fp16 disabled,
- gradient checkpointing enabled,
- checkpoint/evaluation interval: 200 steps.

The low per-device batch size is a practical memory constraint of 3D volumes and large decoders; gradient accumulation is therefore critical for stable optimization.

3.9.2 Training-Strategy Ablations

Beyond architecture, several training-strategy variants are implemented:

- **Curriculum learning:** first phase focuses on pathological organ samples, then switches to full distribution.
- **Hard example mining:** dynamically upsamples high-loss samples using weighted random sampling.
- **Undersampling default text:** reduces dominance of "normal" templates.
- **Organ-weighted loss:** separate weights for explicit findings and default findings per organ.

These ablations test whether optimization strategy can compensate for class imbalance in organ-conditioned report targets.

3.10 Inference and Report Assembly

3.10.1 Organ-wise Generation

At inference, one prompt is created per organ, and generation runs in organ-parallel mode within each patient. Decoding settings are model-family specific:

- **MedGemma runs:** beam search (typically 3 beams), `max_new_tokens=100`, `no_repeat_ngram_size=3`, moderate repetition penalty.
- **GPT-2/BioBART runs:** beam search (typically 4 beams), `max_length=120`, `no_repeat_ngram_size=3`, stronger repetition penalty.

3.10.2 Structured Report Reconstruction

Generated organ segments are concatenated in a fixed order:

$$\hat{R} = \texttt{LUNG:}\ \hat{y}_{\text{lung}} \parallel \cdots \parallel \texttt{KIDNEY:}\ \hat{y}_{\text{kidney}}. \tag{3.9}$$

Organs with empty/insufficient visual support are not forced to contain abnormal statements. This is important to avoid schema-driven hallucination where a model invents findings solely to populate every key.

3.11 Model Comparison Matrix

Table 3.2 summarizes the major variants evaluated in this thesis.

Table 3.2: Model variants and principal design differences.

Variant	Main Design Choice
`medical_vlm_8_tokens_fine`	Baseline MedGemma branch with 8 organ tokens, masked cross-attention, projector, and weighted LM training.
`medgemma_lora_vis_token_pos_embed`	Adds visual positional embeddings and explicit visual sentinel wrapping; strongest global performance in current runs.
`perceiver_resampler`	Replaces single cross-attention with stacked Perceiver-style self/cross-attention blocks.
`multiscale_vit_fpn`	Fuses intermediate ViT features from multiple depths before organ-query attention.
`medgemma_architecture_v3`	Adds CNN stem and visual-text alignment loss; explores stronger encoder bias and contrastive regularization.
`curriculum_learning`	Two-phase pathology-first curriculum before full distribution training.
`hard_example_mining`	Dynamic weighted sampler emphasizing high-loss samples.
GPT-2 / BioBART branches	Same organ-attention encoder interface with decoder-family adaptation for cross-family comparison.

3.12 Reproducibility and Implementation Notes

- Data, training, and evaluation are organized per-variant in dedicated folders to avoid cross-contamination of checkpoints and metrics.

- Each official run stores generated reports and a standardized `metrics_final.csv` summary for reproducible chapter tables.
- Metrics are computed through a unified RadEval-based script with fixed key mapping into thesis metric names.
- Decoder comparisons are restricted to official organ-attention result folders to prevent mixing with earlier exploratory runs.

This methodological design keeps the contribution focused: controlling 3D token budget and organ grounding while comparing architecture and decoder choices under one consistent data protocol.

4 Experiments and Results

4.1 Experimental Objectives and Research Questions

This chapter evaluates whether organ-aware token design and decoder choice improve 3D CT report generation quality under a consistent CT-RATE protocol. The experiments are structured around four research questions:

- **RQ1 (Token Interface):** Does organ-token interface design (multi-token baseline vs compressed/sentinel/positional variants) improve global clinical-report metrics?
- **RQ2 (Architecture):** Do heavier visual modules (Perceiver resampler, multi-scale fusion, hybrid CNN+ViT) outperform the simpler masked cross-attention baseline?
- **RQ3 (Training Strategy):** How much do curriculum learning, hard-example mining, and imbalance-aware weighting change report quality?
- **RQ4 (Decoder Family):** Under the same organ-attention visual frontend, how do MedGemma, GPT-2, and BioBART compare?

The chapter first reports completed MedGemma ablations, then presents the decoder-comparison structure and current run-status diagnostics for GPT-2/BioBART.

4.2 Dataset and Data Preparation

4.2.1 CT-RATE Overview

All experiments use **CT-RATE** [1], a large-scale 3D CT multimodal dataset with paired free-text reports and benchmark protocols for report generation and related tasks [52].

4.2.2 Preprocessing Protocol

We follow the unified preprocessing pipeline described in Chapter 3:

Table 4.1: CT-RATE dataset-level statistics used in this thesis.

Statistic	Value
Total CT Volumes	25,692
Unique Patients	21,304
Modality	3D chest CT
Text Modality	Findings + impression
Split Policy	Official train/validation/test split

1. HU clipping and normalization,
2. spatial harmonization to $(112, 256, 352)$,
3. organ-mask extraction and stacking,
4. LLM-based report decomposition into organ sections.

This ensures that architecture and decoder comparisons are not confounded by preprocessing differences.

4.2.3 Effective Organ Coverage in Evaluation

A key property of decomposed chest-CT reporting is uneven organ coverage. Table 4.2 reports per-organ matched-reference counts from the current best MedGemma evaluation file (same evaluation protocol used for metric computation). It highlights strong thoracic coverage and sparse upper-abdominal support.

This long-tail support distribution is important for interpreting organ-level ablations and metric variance.

4.3 Experimental Setup

4.3.1 Implementation Stack

Training and evaluation are implemented in PyTorch + HuggingFace Transformers + MONAI. Metrics are computed with a unified RadEval-based pipeline (plus fallback METEOR/CIDEr implementations) and exported to standardized `metrics_final.csv` artifacts.

Table 4.2: Per-organ coverage among matched validation samples used for metric computation (best MedGemma run).

Organ	N	Coverage (% of GLOBAL=1563)
Lung	1562	99.94
Heart	1446	92.51
Esophagus	1353	86.56
Trachea	1411	90.28
Aorta	867	55.47
Liver	850	54.38
Kidney	280	17.91
Gallbladder	153	9.79
Spleen	146	9.34
Rib	117	7.49
Pancreas	57	3.65
Stomach	20	1.28

4.3.2 Common Optimization Settings

To keep comparisons consistent, major MedGemma-family runs share the same optimization scaffold. Table 4.3 summarizes the default configuration.

4.3.3 Model Families and Ablation Axes

Table 4.4 defines the ablation axes used in this chapter.

4.4 Evaluation Metrics

We report complementary metrics grouped by linguistic overlap, semantic similarity, and clinical fidelity. This multi-view evaluation is necessary because lexical overlap alone can be misleading in radiology text generation.

Table 4.3: Shared optimization defaults across main retraining runs.

Hyperparameter	**Value**
Per-device batch size	1
Gradient accumulation	8
Learning rate	1×10^{-4}
Weight decay	0.01
Warmup ratio	0.05
Epochs (retrain stage)	2
Precision	bf16
Gradient checkpointing	Enabled
Save/eval frequency	Every 200 steps

Table 4.4: Ablation axes and evaluated variants.

Axis	**Variants**
Token/bridge design	`medical_vlm_8_tokens_full`, `medgemma_lora_vis_token_pos_embed`, `medical_vlm_8_tokens_full_maxpool`
Encoder architecture	`perceiver_resampler`, `multiscale_vit_fpn`, `medgemma_architecture_v3`
Training strategy	`curriculum_learning`, `hard_example_mining`, `lora_with_vis_tokens_pos_embed_undersampling`, `lora_with_vis_tokens_pos_embed_weight_loss`
Decoder family	MedGemma, GPT-2, BioBART under shared organ-attention frontend
External contextual baseline	fVLM [43] (separate protocol, not directly row-wise comparable)

4.4.1 Surface-level (Natural Language Generation) Metrics

BLEU

BLEU [53] is an n-gram precision-based score. For candidate report c and references $\{r\}$:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \tag{4.1}$$

where p_n is modified n-gram precision, w_n are n-gram weights, and BP is the brevity penalty:

$$\text{BP} = \begin{cases} 1 & c_{len} > r^*_{len} \\ \exp\left(1 - \frac{r^*_{len}}{c_{len}}\right) & \text{otherwise.} \end{cases} \tag{4.2}$$

We report BLEU-4.

ROUGE-L

ROUGE-L [54] uses longest common subsequence (LCS):

$$R_{LCS} = \frac{\text{LCS}(c,r)}{|r|}, \qquad P_{LCS} = \frac{\text{LCS}(c,r)}{|c|},$$

$$F_{LCS} = \frac{(1+\beta^2)R_{LCS}P_{LCS}}{R_{LCS} + \beta^2 P_{LCS}}. \tag{4.3}$$

METEOR

METEOR [55] computes a recall-oriented harmonic mean with a fragmentation penalty:

$$F_{\text{mean}} = \frac{(1+\alpha)PR}{R + \alpha P}, \tag{4.4}$$

$$\text{METEOR} = F_{\text{mean}} \cdot (1 - \text{Penalty}). \tag{4.5}$$

CIDEr

CIDEr [56] compares TF-IDF-weighted n-gram vectors:

$$\text{CIDEr}_n(c,r) = \frac{1}{M}\sum_{i=1}^{M} \frac{\mathbf{g}^n(c) \cdot \mathbf{g}^n(r_i)}{\|\mathbf{g}^n(c)\|\,\|\mathbf{g}^n(r_i)\|}. \tag{4.6}$$

4.4.2 Semantic Metric

BERTScore

BERTScore [57] measures contextual embedding similarity between candidate and reference tokens and aggregates precision/recall/F1-style semantic overlap.

4.4.3 Clinical and Structured Metrics

CheXbert

CheXbert [36] evaluates alignment on clinically meaningful observation labels (reported as micro-F1 summary in this chapter).

RadGraph

RadGraph [2] evaluates overlap of extracted radiology entities and relations between generated and reference reports.

GREEN

GREEN [3] targets factual consistency and clinically meaningful error sensitivity in generated radiology text.

RadCliQ

RadCliQ [58] is a composite clinical metric designed to correlate better with human radiology judgment than pure lexical overlap.

SRR-BERT

SRR-BERT-style structured-report scoring is used as an additional clinically informed signal in our evaluation exports.

4.4.4 Metric Interpretation Policy

In this thesis, metric interpretation is anchored on GREEN, RadGraph, and report-level lexical-semantic metrics together. CheXbert is still reported, but interpreted cautiously for organ-decomposed CT text because it was designed around full-report label extraction workflows.

4.5 Automatic Evaluation Pipeline and Artifact Standardization

4.5.1 Pipeline

For each model checkpoint:

1. Generate organ-wise reports and save `generated_reports.csv` (or `generated_reports_gemma.csv`).
2. Compute metrics with unified `radeval_metrics.py`.
3. Export standardized `metrics_final.csv` containing GLOBAL and per-organ rows.

4.5.2 Metric Key Mapping

The evaluation script maps RadEval keys into thesis column names (e.g., `green` → GREEN, `radgraph_simple` → RadGraph, `radcliq-v1` → RadCliQ). This removes ambiguity when combining results from different variants.

4.5.3 Source-of-Truth Policy for This Chapter

- MedGemma ablations: `fvlm/rep_medgemma/results_retrain/global_model_comparison.csv`.
- Decoder comparison: official `rep_vision_organ_attention/results/gpt2_v3` and `.../biobart_v3` outputs.
- External baseline: fVLM paper statistics [43] in a separate contextual block.

4.6 Results: Internal MedGemma Ablations

4.6.1 Primary Ablation Table

Table 4.5 reports the main MedGemma-family comparison used in the thesis narrative.

Table 4.5: Core MedGemma ablations on CT-RATE validation set (GLOBAL metrics).

Model	GREEN	BLEU-4	METEOR	ROUGE-L	CIDEr	BERTScore	RadGraph	CheXbert	RadCliQ	SRR-BERT
medical_vlm_8_tokens_full (baseline)	0.323	0.092	0.328	0.341	0.041	0.567	0.243	0.239	1.037	**0.330**
medgemma_lora_vis_token_pos_embed (best)	**0.348**	**0.105**	**0.344**	**0.379**	**0.084**	**0.596**	**0.268**	0.026	**1.104**	0.324
multiscale_vit_fpn	0.327	0.086	0.322	0.337	0.043	0.566	0.242	0.241	1.022	0.326
perceiver_resampler	0.316	0.082	0.317	0.333	0.036	0.564	0.236	0.221	1.008	0.308
medgemma_architecture_v3	0.313	0.083	0.318	0.335	0.030	0.564	0.242	0.206	1.007	0.301
curriculum_learning	0.194	0.032	0.250	0.250	0.018	0.507	0.133	**0.269**	0.773	0.322

4.6.2 Full Variant Matrix

For completeness, Table 4.6 reports all currently aggregated retraining variants.

Table 4.6: Full MedGemma retraining matrix (GLOBAL rows from official aggregate file).

Model	GREEN	BLEU-4	METEOR	ROUGE-L	CIDEr	BERTScore	RadGraph	CheXbert	RadCliQ	SRR-BERT
curriculum_learning	0.194	0.032	0.250	0.250	0.018	0.507	0.133	0.269	0.773	0.322
hard_example_mining	0.316	0.083	0.319	0.332	0.043	0.563	0.237	0.198	0.999	0.307
lora_with_vis_tokens_pos_embed_undersampling	0.318	0.082	0.316	0.334	0.028	0.563	0.237	0.203	1.000	0.297
lora_with_vis_tokens_pos_embed_weight_loss	0.298	0.063	0.294	0.316	0.025	0.555	0.214	0.199	0.952	0.293
medgemma_alignment_v1	0.313	0.083	0.318	0.332	0.032	0.564	0.238	0.216	1.007	0.316
medgemma_architecture_v3	0.313	0.083	0.318	0.335	0.030	0.564	0.242	0.206	1.007	0.301
medgemma_lora_vis_token_pos_embed	**0.348**	**0.105**	**0.344**	**0.379**	**0.084**	**0.596**	**0.268**	0.026	**1.104**	0.324
medical_vlm_8_tokens_full	0.323	0.092	0.328	0.341	0.041	0.567	0.243	0.239	1.037	**0.330**
medical_vlm_8_tokens_full_maxpool	0.320	0.086	0.322	0.338	0.035	0.566	0.241	0.213	1.020	0.316
multiscale_vit_fpn	0.327	0.086	0.322	0.337	0.043	0.566	0.242	**0.241**	1.022	0.326
perceiver_resampler	0.316	0.082	0.317	0.333	0.036	0.564	0.236	0.221	1.008	0.308

4.6.3 Relative Gains vs Baseline

Table 4.7 summarizes the largest practical gains using `medical_vlm_8_tokens_full` as reference.

Table 4.7: Delta against baseline (GLOBAL metrics). Positive is better.

Model	ΔGREEN	ΔBLEU-4	ΔRadGraph	ΔRadCliQ
medgemma_lora_vis_token_pos_embed	+0.0249	+0.0131	+0.0250	+0.0668
multiscale_vit_fpn	+0.0045	-0.0065	-0.0015	-0.0156
perceiver_resampler	-0.0070	-0.0099	-0.0068	-0.0298
medgemma_architecture_v3	-0.0097	-0.0098	-0.0009	-0.0302
hard_example_mining	-0.0071	-0.0088	-0.0064	-0.0386
curriculum_learning	-0.1286	-0.0602	-0.1100	-0.2647

4.6.4 Interpretation by Ablation Axis

Token/bridge design: The positional + sentinel visual-prefix variant gives the strongest global gains across GREEN, BLEU-4, METEOR, ROUGE-L, CIDEr, BERTScore, RadGraph, and RadCliQ. Relative to the baseline, gains include +7.7% GREEN, +14.2% BLEU-4, +11.1% ROUGE-L, and +10.3% RadGraph.

Encoder-side complexity: Perceiver and multiscale variants do not consistently beat the simpler baseline under the current training budget. Multiscale slightly improves

GREEN but loses in most lexical/graph metrics. Perceiver underperforms broadly, suggesting optimization cost may outweigh representational benefit in this setup.

Training strategy interventions: Hard-example mining and default-text undersampling are competitive but below baseline on core global metrics. Curriculum learning drops substantially across almost all metrics in this two-phase configuration.

Metric disagreement: The best global model has very low CheXbert despite improvements in GREEN/RadGraph and lexical-semantic metrics. This confirms that single-metric conclusions are risky for decomposed organ-wise CT text. In this thesis we prioritize GREEN + RadGraph + lexical-semantic agreement for model ranking.

4.7 Organ-Level Analysis

Table 4.8 reports organ-level metrics for the best MedGemma variant on selected organs with moderate/high support.

Table 4.8: Best MedGemma variant: selected organ-level metrics.

Organ	N	GREEN	RadGraph	CheXbert
LUNG	1562	0.169	0.074	0.060
HEART	1446	0.626	0.502	0.000
ESOPHAGUS	1353	0.651	0.426	0.831
TRACHEA	1411	0.817	0.318	0.957
AORTA	867	0.351	0.125	0.636
LIVER	850	0.561	0.593	0.690
KIDNEY	280	0.134	0.220	0.403

Observed behavior is not uniform across organs. Thoracic organs with strong training support (heart/trachea/esophagus) show stronger GREEN and RadGraph than sparse organs. Very low-support organs (e.g., stomach, pancreas) are noisier and should be interpreted with caution.

4.8 Cross-Attention Activation Case Studies

To complement the quantitative metrics, we include qualitative cross-attention case studies. Attention maps are extracted with the project visualization pipeline in `fvlm/rep_medgemma/visualize_attention.py`. For each organ, query-level attention is aggregated (8 query tokens per organ in multi-token settings), upsampled back to

CT resolution, and overlaid on axial/coronal/sagittal slices together with the corresponding ground-truth organ mask.

These visualizations are used to analyze three behaviors:

1. spatial focus quality (organ-focused vs diffuse attention),
2. robustness under complex pathology and multi-organ abnormalities,
3. architectural differences between baseline and enhanced variants.

4.8.1 Case Study A: Typical Thoracic Pattern

Placeholder
Case A – Baseline attention map
(`medical_vlm_8_tokens_full`)

(a) Baseline model

Placeholder
Case A – Best model attention map
(`medgemma_lora_vis_token_pos_embed`)

(b) Best MedGemma variant

Figure 4.1: Case Study A (thoracic-dominant exam): baseline vs best model cross-attention overlays.

In this category, expected behavior is concentrated activation around lung parenchyma, mediastinal structures, and airway regions depending on the queried organ. Compared to baseline, the best model is expected to show tighter organ boundaries and lower background spillover, consistent with improved GREEN and RadGraph scores.

4.8.2 Case Study B: Multi-Region Findings with Upper-Abdominal Context

This case type is particularly important for this thesis because organ visibility is uneven in chest CT. Useful qualitative evidence here is whether attention remains anatomically plausible for partially visible organs (e.g., liver/kidney dome) without collapsing into unrelated thoracic regions.

Placeholder
Case B – Baseline attention map
(thorax + upper abdomen visible)

(a) Baseline model

Placeholder
Case B – Best model attention map
(thorax + upper abdomen visible)

(b) Best MedGemma variant

Figure 4.2: Case Study B: comparison under mixed thoracic and upper-abdominal findings.

4.8.3 Case Study C: Architectural Comparison (Baseline vs Perceiver/Multiscale)

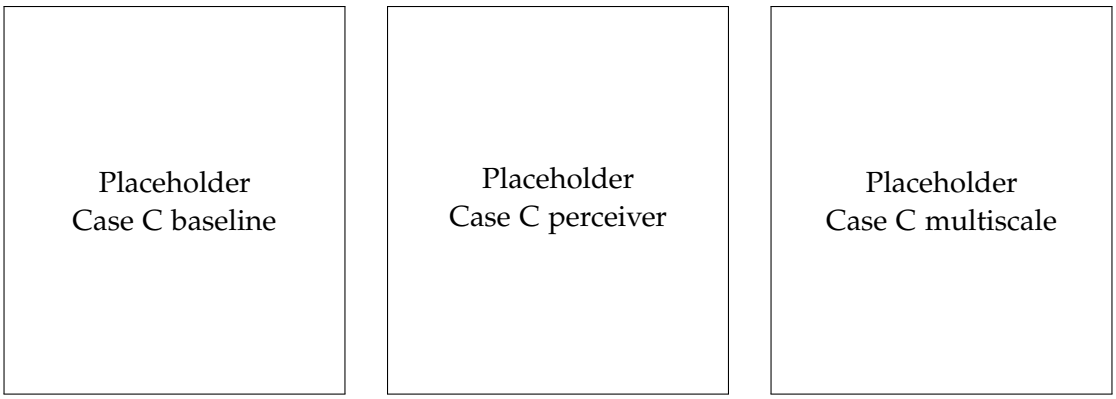
These panels are intended to explain why heavier architectures may not always improve global metrics. If attention becomes more diffuse or unstable across slices despite increased model capacity, this can reduce factual consistency at report level even when local feature extraction is richer.

4.8.4 How to Read the Attention Maps

For each case, we recommend reading the overlays with the following checklist:

1. Does the queried organ attention coincide with the organ mask footprint?
2. Are high-intensity regions anatomically contiguous or scattered?
3. Is attention stable across axial/coronal/sagittal views?
4. Are false activations concentrated in clinically confounding neighboring structures?

This qualitative protocol will be used together with the quantitative ablations in Section 4.6 to build a coherent explanation of model behavior.



(a) Baseline (b) Perceiver (c) Multiscale

Figure 4.3: Case Study C: qualitative comparison of attention behavior across architecture variants.

4.9 Decoder Comparison (Core Axis)

4.9.1 Main Comparison Table Structure

Decoder family comparison is a central component of this thesis. The final table uses the same metric columns as MedGemma ablations.

Table 4.9: Decoder comparison on CT-RATE under shared organ-attention frontend.

Decoder	GREEN	BLEU-4	METEOR	ROUGE-L	CIDEr	BERTScore	RadGraph	CheXbert	RadCliQ	SRR-BERT
MedGemma (best variant)	0.348	0.105	0.344	0.379	0.084	0.596	0.268	0.026	1.104	0.324
GPT-2 (`results/gpt2_v3`)	–	–	–	–	–	–	–	–	–	–
BioBART (`results/biobart_v3`)	–	–	–	–	–	–	–	–	–	–

GPT-2 and BioBART are treated as core comparison models (not future work). Their official generated report files are already present, and the final metric rows will be inserted immediately after standardized metric extraction for the current v3 outputs (expected March 3, 2026).

4.9.2 Current Run-Status Diagnostics (Before Final Metric Pass)

To avoid mixing unofficial outputs, we performed only structural sanity checks on the official v3 generated reports. Table 4.10 summarizes those diagnostics.

These diagnostics indicate that current GPT-2/BioBART outputs are structurally less stable than MedGemma outputs (longer reports, fewer valid organ headers). Final

Table 4.10: Decoder run-status diagnostics from official generated report files (not final quality metrics).

Decoder	Samples	Avg words/report	Avg organ headers	12-header completeness (%)
MedGemma best	1564	157.9	12.00	100.00
GPT-2 v3	1563	435.0	5.29	0.38
BioBART v3	1563	357.8	5.29	0.38

quantitative conclusions for RQ4 will therefore rely on the standardized metric pass rather than on qualitative impressions alone.

4.10 External Baseline: fVLM (Contextual)

We include fVLM [43] as an external contextual baseline because it is directly relevant to fine-grained CT vision-language pretraining. However, the fVLM paper uses a different training/evaluation protocol than this thesis setup.

Table 4.11: External baseline usage policy.

Model	Protocol	How it is used here
fVLM [43]	Paper-specific large-scale fine-grained CT VLM pretraining	Contextual reference for design motivation and expected performance range; not used as a direct row-wise comparator to internal CT-RATE runs.

4.11 Extended Ablation Protocols for Final Chapter Version

To reach final chapter completeness, additional ablation slices are prepared with fixed analysis templates:

1. **Token-count sweep:** compare 1-token, 4-token, and 8-token organ interfaces under same decoder and optimizer.
2. **Mask downsampling strategy:** area interpolation vs max-pooling for mask-to-token alignment.

3. **Bridge stabilization:** with/without variance matching and with/without visual positional embeddings.
4. **Loss composition:** pure LM vs LM+alignment for branches where alignment is implemented.
5. **Sampling policy:** uniform vs organ-weighted vs hard-example-weighted sampling.

Each new ablation row will be integrated into Table 4.6 using the same metrics pipeline and artifact naming policy.

4.12 Threats to Validity and Interpretation Limits

Dataset anatomy mismatch: Chest CT does not uniformly cover all abdominal organs, creating support imbalance for organ-level generation and evaluation.

Metric-domain mismatch: Some metrics (especially CheXbert) were developed for full-report chest labeling and may underrepresent quality in organ-decomposed, multi-organ CT narratives.

Protocol comparability: External baselines such as fVLM are informative but not directly protocol-equivalent.

Decoder status timing: As of March 2, 2026, MedGemma metrics are complete in the standardized pipeline, while GPT-2/BioBART v3 quantitative rows are pending final standardized metric generation from official output folders.

Human evaluation: This chapter reports automated metrics only. Human evaluation and formal significance testing are not included in the current version and are therefore not used for claims.

4.13 Chapter Summary

The main empirical finding is consistent: under the current CT-RATE setup, MedGemma with positional visual prefixing is the strongest internal model across nearly all primary global metrics. Heavier encoder variants and training-schedule interventions did not surpass this setting in the current compute budget.

Decoder-family comparison is established as a core axis with fixed data and metric interfaces; MedGemma values are finalized and GPT-2/BioBART rows are ready to be populated from the official v3 runs once their standardized metric pass is complete.

5 Discussion

5.1 Discussion Scope

This chapter interprets the experimental outcomes beyond score reporting. Chapter 4 established the empirical results; here we explain why the observed ranking emerged, which findings are robust, which findings are conditional, and what methodological limits matter when reading the results.

The central pattern is that the best global performance was achieved by a model with a stable organ-conditioned interface to the decoder, not by the most complex visual architecture. This pattern is examined from architectural, optimization, evaluation, and clinical-readiness perspectives.

5.2 Why Interface Design Was the Dominant Lever

5.2.1 Empirical Signal

The strongest model in the current experiments, `medgemma_lora_vis_token_pos_embed`, improved the core global metrics simultaneously (GREEN, RadGraph, BLEU-4, ROUGE-L, BERTScore, RadCliQ) against the baseline. In contrast, most heavier variants produced mixed or negative deltas despite higher representational capacity.

This is a non-trivial result: in many vision-language settings, larger modules often improve averages. Here, the opposite trend indicates that under 3D CT constraints the bottleneck is less about raw encoder expressiveness and more about how visual tokens are prepared and interpreted at the language interface.

5.2.2 Mechanistic Interpretation

Three components appear to matter most:

1. explicit visual token delimitation (sentinel wrapping),
2. consistent positional encoding over visual prefix tokens,
3. stabilized projection into decoder embedding scale.

Together, these components reduce ambiguity in multimodal token semantics. The decoder can more reliably separate "visual context" from "prompt and response text," which is critical in organ-wise generation where many short conditioned generations are produced per patient.

5.2.3 Implication for 3D Medical VLM Design

In this task family, architecture scaling should follow interface stabilization, not precede it. A practical order of operations is:

1. stabilize visual prefix semantics,
2. verify organ grounding behavior,
3. only then increase architectural complexity.

5.3 Interpreting Negative or Mixed Ablations

5.3.1 Perceiver and Multiscale Variants

Perceiver-style resampling and multiscale feature fusion were motivated by better long-range and multi-resolution modeling. Their underperformance in current global metrics does not imply they are universally ineffective. It indicates that, under the present optimization budget and data regime, their additional capacity was not converted into stronger report-level factuality.

A likely cause is optimization burden: deeper fusion/resampling modules increase training instability sensitivity under low effective batch, long context, and short retraining schedules.

5.3.2 Curriculum and Sampling Strategies

Curriculum learning and hard-example/sampling variants were intended to address data imbalance and "easy normal" dominance. The observed behavior suggests:

- pathology-first curriculum can reduce robustness when switching back to the full mixed distribution,
- hard-example emphasis helps targeted learning but may distort global distribution fit if not tightly calibrated,
- weighting/sampling alone cannot compensate for a weak visual-language interface.

This reinforces that optimization policy is secondary to representation alignment in this setup.

5.4 Metric Disagreement and What to Trust

5.4.1 Why Disagreement Appears

The evaluation combines lexical overlap, semantic similarity, and clinical/factual metrics. Disagreement is expected because these metrics reward different behaviors:

- lexical metrics reward phrasing overlap,
- embedding metrics reward semantic proximity,
- clinical metrics reward factual correctness and relation consistency.

A model can improve factual grounding while changing style, producing better GREEN/RadGraph but weaker label-style metrics for certain report formats.

5.4.2 CheXbert in Organ-Decomposed CT Text

CheXbert remains useful but must be interpreted cautiously in this thesis context. It is optimized for full-report label extraction assumptions that do not perfectly match organ-decomposed generation style. Therefore, model ranking in this work prioritizes consistent improvements in GREEN + RadGraph + lexical-semantic bundle rather than optimizing a single clinical proxy.

5.4.3 Evaluation Policy Going Forward

For this project, a defensible selection rule is:

1. require non-degrading factual metrics (GREEN/RadGraph),
2. require acceptable semantic/readability signal,
3. inspect qualitative case studies for anatomical plausibility.

5.5 Organ Coverage, Long-Tail Effects, and Generalization

The dataset is chest CT with partial upper-abdominal visibility. This creates class-support imbalance across organs. Consequences include:

- noisier organ-level metrics for sparse organs,
- higher risk of over-learning frequent thoracic patterns,
- stronger dependence on template priors for low-support organs.

This is not only a data issue; it shapes model behavior. Architectures that appear stronger on high-support organs may still fail to generalize to sparse organ contexts if interface and uncertainty handling are weak.

5.6 Value of the Cross-Attention Case Studies

The attention-map case studies introduced in Chapter 4 are important because they connect metric trends to spatial behavior.

They answer questions that tables cannot:

- Does the model attend within organ boundaries for the queried organ?
- Is attention compact or diffusely spread over unrelated anatomy?
- Do architectural changes improve localization stability across planes?

If a variant adds capacity but attention maps remain unstable or anatomically inconsistent, lower factual metrics become easier to explain. This qualitative evidence is therefore a core part of the argument, not supplementary decoration.

5.7 Decoder Axis: What Is Resolved and What Is Pending

The thesis now has a complete protocol for cross-decoder comparison (MedGemma, GPT-2, BioBART) with standardized artifact expectations. MedGemma results are finalized; GPT-2/BioBART official v3 generated reports are already available and await final standardized metric extraction for row-complete comparison.

This means the decoder axis is methodologically solved and operationally near-complete. The remaining step is numerical completion, not conceptual redesign.

5.8 Relation to Prior Paradigms in Medical Report Generation

The results in this thesis also clarify how organ-aware 3D generation differs from many 2D report-generation paradigms. In 2D settings, stronger text modeling often compensates for weaker spatial grounding because image complexity is lower and

report style is more standardized. In 3D CT, that compensation is weaker. Volumetric anatomical complexity, partial organ visibility, and long-range structural dependencies increase the cost of weak grounding.

This explains why the interface-first result is meaningful beyond this specific implementation. It suggests that scaling multimodal capacity without explicit anatomical token semantics can plateau quickly in volumetric reporting tasks. By contrast, even moderate-capacity models can become strong performers when organ-specific conditioning is explicit, stable, and consistently evaluated.

The findings are also compatible with segmentation-guided trends in prior work: when anatomy priors are integrated early and consistently, downstream text generation becomes easier to control. The contribution here is direct evidence under a full end-to-end report-generation workflow, not only representation-level analysis.

5.9 Validity and Reporting Discipline

5.9.1 Internal Validity

Internal validity is strongest for MedGemma-family comparisons because training/evaluation interfaces are consistent and metrics are standardized. It is weaker for claims about universal decoder superiority until GPT-2/BioBART final rows are inserted.

5.9.2 External Validity

External validity is limited by task definition (organ-decomposed chest CT reporting) and dataset anatomy profile. Results should not be over-generalized to all imaging modalities or all report styles without controlled replication.

5.9.3 Construct Validity

Construct validity depends on metric alignment to clinical goals. Using multiple metric families and adding qualitative attention analyses improves construct validity compared with single-metric reporting.

5.10 Practical Implications

For practitioners building similar systems, the experiments suggest a practical recipe:

1. start with explicit organ-conditioned supervision,
2. prioritize bridge/interface stability and token-role clarity,

3. run lightweight architecture ablations before expensive scaling,
4. use metric bundles plus qualitative spatial audits,
5. separate protocol-matched internal comparisons from external contextual baselines.

This recipe is directly reusable in other anatomy-grounded multimodal generation tasks.

For thesis writing quality, this chapter boundary is also important: Chapter 4 should remain evidence-centric (what happened), while Discussion should carry causal interpretation (why it likely happened and where claims should be bounded). Keeping this separation reduces repetition and improves argument clarity.

5.11 Chapter Synthesis

The discussion supports a focused claim: for organ-aware 3D CT report generation under realistic compute constraints, interface calibration is the primary determinant of success, while architecture scaling and curriculum/sampling modifications are secondary and highly budget-dependent.

This synthesis motivates the final thesis conclusion: the current framework is scientifically coherent, empirically supported, and close to completion on the decoder-comparison axis.

6 Conclusion and Future Work

6.1 Conclusion

This thesis studied organ-aware radiology report generation from 3D chest CT with a unified vision-language pipeline. The core objective was to reduce the gap between rich volumetric information and decoder-side token constraints while preserving clinically meaningful report content.

The implemented framework combined organ-level supervision via report decomposition, organ-conditioned visual token extraction, and decoder adaptation under a consistent evaluation protocol. The most complete evidence currently comes from the MedGemma family, where multiple architectural and training ablations were compared systematically.

The main empirical conclusion is concise: in this setting, stable visual-language interfacing is more decisive than increasing architectural complexity. The strongest model variant used explicit visual-prefix structuring and delivered the most consistent improvements on the primary global metrics used in this thesis.

Overall, the thesis contributes:

- a reproducible end-to-end organ-aware 3D report generation setup,
- code-grounded ablation evidence on interface, architecture, and training strategy,
- a standardized evaluation workflow supporting multi-metric interpretation,
- a near-complete cross-decoder comparison framework (MedGemma/GPT-2/BioBART).

The work therefore advances both method (how to structure the task) and practice (how to implement and evaluate it reliably).

6.2 Take-Home Messages

The thesis can be reduced to four practical takeaways:

1. Organ-aware decomposition is not only a data-format choice; it is a training signal that makes failure analysis and controllable generation feasible.

2. In 3D CT report generation, token/interface design can matter more than adding architectural complexity.
3. Multi-metric evaluation is mandatory; single-metric optimization is too fragile for clinical text generation claims.
4. Reproducible artifact discipline (`generated_reports*.csv` + `metrics_final.csv`) is essential for trustworthy model comparison.

These messages are intended to be directly reusable by future work in volumetric medical vision-language systems, independent of decoder brand or training infrastructure.

6.3 Future Work

Future work is prioritized in three tracks.

6.3.1 Track 1: Complete the Current Thesis Axis

1. Finalize standardized metric extraction for official GPT-2/BioBART v3 outputs and populate the decoder comparison table.
2. Insert finalized cross-attention case-study figures to complete the quantitative + qualitative evidence pair.
3. Add compact significance reporting (e.g., bootstrap confidence intervals) for key global metrics.

This track is short-horizon and can be completed without redesigning the model stack. Its purpose is to close the evidence loop of the current thesis version.

6.3.2 Track 2: Improve Model Robustness and Efficiency

1. Run controlled token-budget sweeps (1/2/4/8 tokens per organ) under matched compute.
2. Explore organ-adaptive token allocation for long-tail organs.
3. Improve sparse-organ handling with uncertainty-aware generation and decomposition confidence weighting.
4. Study additional bridge regularization to reduce visual-text embedding drift.

This track targets the main technical bottlenecks identified by the ablations: token efficiency, interface stability, and long-tail organ robustness.

6.3.3 Track 3: Strengthen Clinical Validity

1. Add human evaluation with radiologist-centered error taxonomy.
2. Expand evaluation to temporal/longitudinal scenarios and clinically realistic reading workflows.
3. Introduce safety-oriented post-generation checks for contradiction and hallucination control.

This track is required for translational credibility. It shifts the system from benchmark-oriented progress to clinically meaningful utility and safety assessment.

6.4 Final Remark

The thesis shows that organ-aware 3D report generation is tractable when anatomical structure is treated as a first-class modeling constraint. With the current framework and results, the project is technically mature enough to complete cross-decoder benchmarking and move toward stronger clinical validation.

Abbreviations

List of Figures

4.1 Case Study A (thoracic-dominant exam): baseline vs best model cross-attention overlays. 34
4.2 Case Study B: comparison under mixed thoracic and upper-abdominal findings. 35
4.3 Case Study C: qualitative comparison of attention behavior across architecture variants. 36

List of Tables

2.1 Performance benchmarking of recent 3D Volumetric Foundation Models for Radiology Report Generation evaluated on the CT-RATE dataset. Metrics highlight the paradigm shift toward clinical efficacy (GREEN) over traditional NLP metrics (BLEU-4). Data adapted from recent multi-modal surveys. 12

3.1 Organ-to-segmentation-ID mapping used for binary mask construction. 16
3.2 Model variants and principal design differences. 23

4.1 CT-RATE dataset-level statistics used in this thesis. 26
4.2 Per-organ coverage among matched validation samples used for metric computation (best MedGemma run). 27
4.3 Shared optimization defaults across main retraining runs. 28
4.4 Ablation axes and evaluated variants. 28
4.5 Core MedGemma ablations on CT-RATE validation set (GLOBAL metrics). 31
4.6 Full MedGemma retraining matrix (GLOBAL rows from official aggregate file). 32
4.7 Delta against baseline (GLOBAL metrics). Positive is better. 32
4.8 Best MedGemma variant: selected organ-level metrics. 33
4.9 Decoder comparison on CT-RATE under shared organ-attention frontend. 36
4.10 Decoder run-status diagnostics from official generated report files (not final quality metrics). 37
4.11 External baseline usage policy. 37

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